

UseRotationOfMarker [SimTalk]

Sets if the AGV uses the rotation of the *Marker* designated by <Path> when driving to its destination (true) or not (false).

Type

Attribute

Syntax

```
<Path>.UseRotationOfMarker: boolean
```

Assignment Value

You can assign a value of data type *boolean*.

Example

```
MyMarker.UseRotationOfMarker := true
```

See also

[Amount \[text box\] - AGVPool](#)

ShiftCalendar

ShiftCalendar [object]

Use the object *ShiftCalendar* for modeling the shift system and organize the shift work performed in your plant.

Image



Description

You can define as many shifts in your simulation model as you need. A shift pauses one or several machines in your model, by setting the boolean attributes *Pause* and *Unplanned* of the respective material flow objects to true.

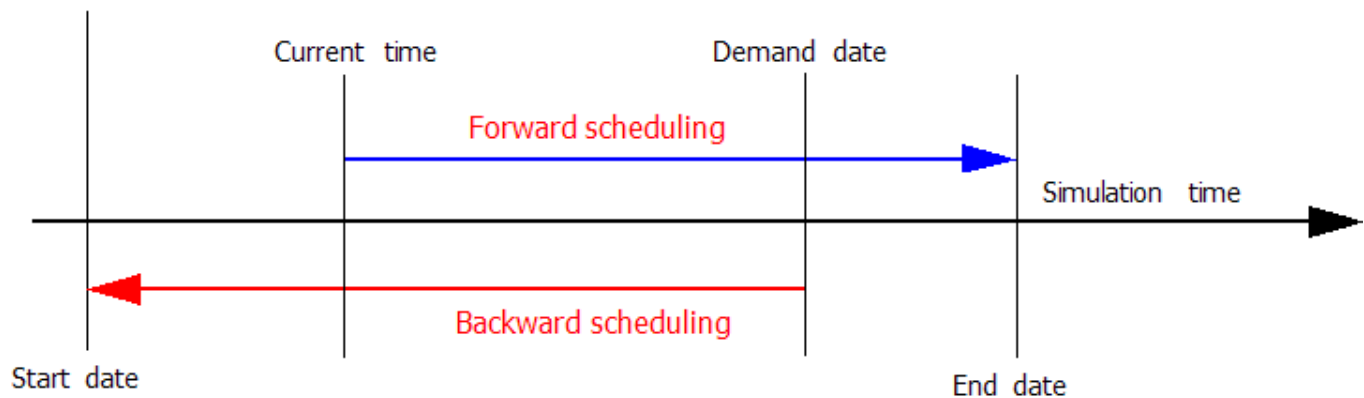
Note

If you deactivate the *Pause* during the unplanned time, either by clearing the check box **Pause** in the dialog of the object, or by setting the attribute *Pause* to `false`, the object starts to work.

Use the method **schedule** to make the *ShiftCalendar* set the date and time to start or to finish the production process. We distinguish between:

- **Forward scheduling** beginning from the start date forward into the future, and
- **Backward scheduling** which computes the start date going backward in time from the demand date.

Normally, you start with the demand date, computing the start date through backward scheduling. If this start date is located in the past though, you have to recompute the dates beginning with the present time and forward-schedule the end date.



To show a **tooltip** with information about the *ShiftCalendar*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *ShiftCalendar*, click **Show Manipulators**



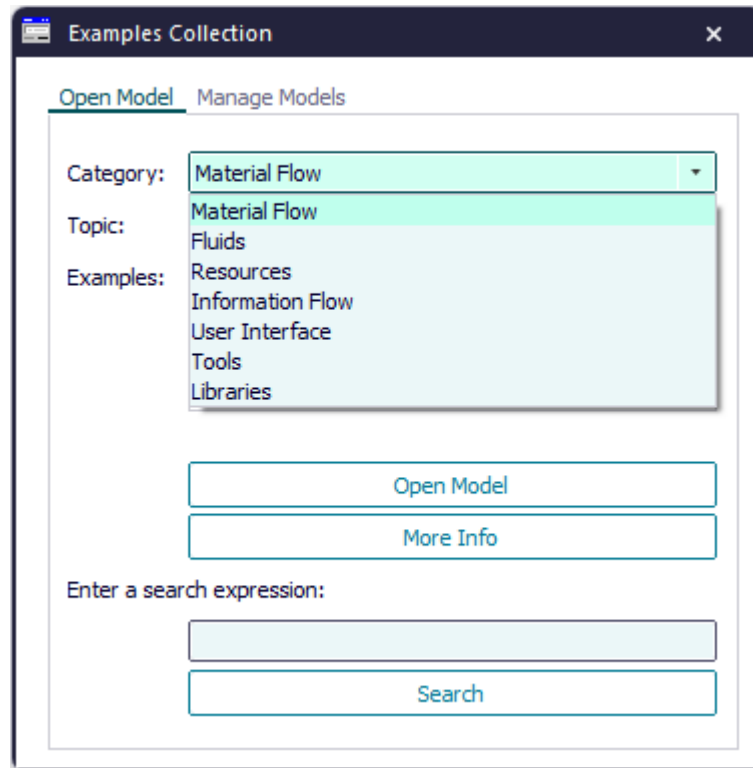
on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model



To add the object *ShiftCalendar* to your simulation model, click **Manage Class Library** > **Basic Objects** > **Resources** > **ShiftCalendar** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Model a Shift System](#)

[Shift Calendar \[tab Controls\]](#)

[Select Shift Calendar](#)

Dialog Box of the ShiftCalendar

Dialog Box of the ShiftCalendar

Double-click the icon of the *ShiftCalendar* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.

Note

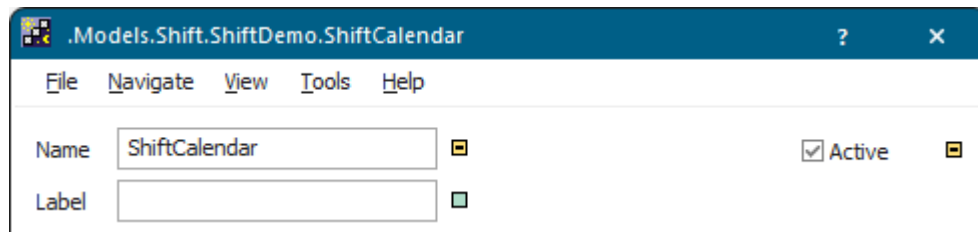
Before you can type in data, click the **Inheritance** check box so that it looks like this .

See also

[Active \[check box\] - ShiftCalendar](#)

Active [check box] - ShiftCalendar

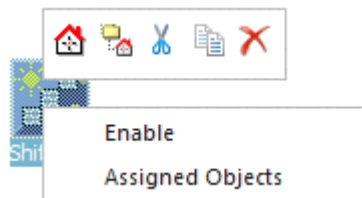
To make your facility work in shifts, select this check box. To deactivate shift work, clear the check box.



Remarks

Instead, you can also right-click the *ShiftCalendar* object in the *Frame* and select **Activate** on the context menu.

To deactivate shift operation, select **Deactivate** on the context menu.



SimTalk

Active [SimTalk] - ShiftCalendar

Tab Shift Times

Tab Shift Times

Type in the name of the **Shift**, the time it starts, the time it ends, and the duration of **Pauses** on the tab **Shift Times**.

Remarks

Note

Before you can type data into the table on the tab, click the **Inheritance** check box so that it is turned off .

To select which shift works on which day, select the respective check box in the cells below **Mo, Tu, We, Th, Fr, Sa, Su**.

Shift Times		Calendar		Resources		User-defined					
	Shift	From	To	Mo	Tu	We	Th	Fr	Sa	Su	Pauses
1	Morning	6:00	14:00	☒	☒	☒	☒	☒	☐	☐	9:00-9:15; 12:00-1
2	Evening	14:00	22:00	☒	☒	☒	☒	☒	☐	☐	18:00-18:30; 20:30
3	Graveyard	22:00	6:00	☐	☐	☐	☐	☒	☐	☐	0:30-0:45; 2:45-3:

To manipulate the contents of the list, use the commands on the [Context Menu of Embedded Lists](#).

SimTalk

[schedule \[SimTalk\] - ShiftCalendar](#)

[ShiftPlan \[SimTalk\]](#)

See also

[Shift \[ShiftCalendar\]](#)

From

To

[Pauses \[ShiftCalendar\]](#)

[Specify the Names of the Shifts, the Respective Times and Days](#)

Shift [ShiftCalendar]

Type in the names of the different shifts into these cells.

Shift Times		Calendar		Resources		User-defined					
	Shift	From	To	Mo	Tu	We	Th	Fr	Sa	Su	Pauses
1	Morning	6:00	14:00	☒	☒	☒	☒	☒	☐	☐	9:00-9:15; 12:00-1
2	Evening	14:00	22:00	☒	☒	☒	☒	☒	☐	☐	18:00-18:30; 20:30
3	Graveyard	22:00	6:00	☐	☐	☐	☐	☒	☐	☐	0:30-0:45; 2:45-3:0

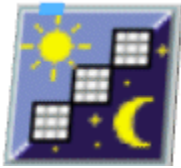
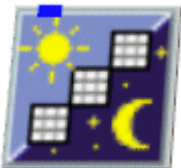
Remarks

Note

Before you can type data into the table on the tab, click the **Inheritance** check box so that it is turned off .

You might, for example, type in Morning, Evening, Graveyard, etc.

The *ShiftCalendar* displays these state graphics:

Unplanned	Planned
	

SimTalk

[GetCurrShift \[SimTalk\]](#)

[ShiftPlan \[SimTalk\]](#)

From

From

Type the times at which the respective shifts start in these cells.

Shift Times		Calendar		Resources		User-defined					
	Shift	From	To	Mo	Tu	We	Th	Fr	Sa	Su	Pauses
1	Morning	6:00	14:00	☒	☒	☒	☒	☒	☐	☐	9:00-9:15; 12:00-1
2	Evening	14:00	22:00	☒	☒	☒	☒	☒	☐	☐	18:00-18:30; 20:30
3	Graveyard	22:00	6:00	☐	☐	☐	☐	☒	☐	☐	0:30-0:45; 2:45-3:1

Remarks

Note

Before you can type data into the table on the tab, click the **Inheritance** check box so that it is turned off .

- Type in a time between 0:00 o'clock and 24:00 o'clock. You can only type in hours and minutes, not hours, minutes and seconds.
- To create a shift located within one day, type in a greater number for the time at which it ends than for the time at which it starts. You might, for example, define a **Morning** shift that starts at 6:00 in the morning and ends at 14:00 on the same day.
 - To create a shift that spans two days, type in a smaller number for the time at which it ends than for the time at which it starts. You might, for example, define a **Graveyard** shift that starts at 22:00 at night on one day, and ends at 6:00 in the morning on the next day.

Note

You cannot enter overlapping shift times.

SimTalk

ShiftPlan [SimTalk]

To

Type the times at which the respective shifts end into these cells.

Shift Times		Calendar		Resources		User-defined					
	Shift	From	To	Mo	Tu	We	Th	Fr	Sa	Su	Pauses
1	Morning	6:00	14:00	☒	☒	☒	☒	☒	☐	☐	9:00-9:15; 12:00-1
2	Evening	14:00	22:00	☒	☒	☒	☒	☒	☐	☐	18:00-18:30; 20:30
3	Graveyard	22:00	6:00	☐	☐	☐	☐	☒	☐	☐	0:30-0:45; 2:45-3:

Remarks

Note

Before you can type data into the table on the tab, click the **Inheritance** check box so that it is turned off .

Type in a time between 0 : 00 o'clock and 24 : 00 o'clock. You can only specify hours and minutes, not hours, minutes and seconds.

- To create a shift located within one day, type in a greater number for the time at which it ends than for the time at which it starts. You might, for example, define a **Morning** shift that starts at 6:00 in the morning and ends at 14:00 the same day.
- To create a shift that spans two days, type in a smaller number for the time at which it ends than for the time at which it starts. You might, for example, define a **Graveyard** shift that starts at 22:00 at night on one day, and ends at 6:00 in the morning on the next day.

Note

You cannot enter overlapping shift times.

SimTalk

ShiftPlan [SimTalk]

Mo, Tu, We, Th, Fr, Sa, Su

Mo, Tu, We, Th, Fr, Sa, Su

To select the individual days on which the shift in the row is active, click the respective check box in the cells below the days of the week.

Shift Times		Calendar		Resources		User-defined					Pauses	
	Shift	From	To	Mo	Tu	We	Th	Fr	Sa	Su		
1	Morning	6:00	14:00	☒	☒	☒	☒	☒	☐	☐	9:00-9:15; 12:00-1	
2	Evening	14:00	22:00	☒	☒	☒	☒	☒	☐	☐	18:00-18:30; 20:30	
3	Graveyard	22:00	6:00	☐	☐	☐	☐	☒	☐	☐	0:30-0:45; 2:45-3:1	

Remarks

Note

Before you can type data into the table on the tab, click the **Inheritance** check box so that it is turned off .

You might, for example:

- Define a **Morning** shift that works **Monday** up to and including **Friday**.
- Define an **Evening** shift that works **Monday** up to and including **Sunday**.

SimTalk

ShiftPlan [SimTalk]

Pauses [ShiftCalendar]

Type in the times of the breaks for each shift into these cells.

Shift Times		Calendar		Resources		User-defined					
	Shift	From	To	Mo	Tu	We	Th	Fr	Sa	Su	Pauses
1	Morning	6:00	14:00	☒	☒	☒	☒	☒	☐	☐	9:00-9:15; 12:00-1
2	Evening	14:00	22:00	☒	☒	☒	☒	☒	☐	☐	18:00-18:30; 20:30
3	Graveyard	22:00	6:00	☐	☐	☐	☐	☒	☐	☐	0:30-0:45; 2:45-3:

Remarks

Note

Before you can type data into the table on the tab, click the **Inheritance** check box so that it is turned off .

Type in the hour and the minute when the break starts, a hyphen and the hour and minute the break ends. When you define several breaks, separate them with a semicolon.

To define a coffee break from 9 o'clock to a quarter past 9 o'clock, and a lunch break from 12 o'clock to a quarter till 1 o'clock, type in 9:00-9:15;12:00-12:45.

To make the *ShiftCalendar* check if the values you typed in for the breaks are plausible or not and if you used the correct format to enter the break times, click **Apply**.

SimTalk

ShiftPlan [SimTalk]

Tab Calendar

Tab Calendar

Define the days on which your plant does not work at all or only works part of the time, and a **Comment** describing the event on the tab **Calendar**.

Remarks

Note

Before you can use the date picker, click the check box **Inheritance** so that it looks like this .

Shift Times				
Calendar				
Resources				
User-defined				
	Date From	Date To	Reduce Time To	Comment
1	2015/05/13		12:00 -24:00	Birthday of Boss
2	2015/05/25			Memorial Day
3	2015/07/03			Independence Day
4	2015/07/13	2015/07/19		Retooling
5	2015/09/07			Labor Day

You can also import a calendar that you saved. Or you can export a calendar you defined and then import it into another simulation model. Click into the list field with the right mouse button and select **Import** or **Export**.

To manipulate the contents of the list, use the commands on the [Context Menu of Embedded Lists](#).

SimTalk

[Calendar \[SimTalk\]](#)

[schedule \[SimTalk\] - ShiftCalendar](#)

See also

[Date From](#)

[Date To](#)

[Reduce Time To](#)

[Comment \[reduce time to\]](#)

[Specify Times During Which the Plant Works Part of the Time](#)

Date From

Type the date at which your plant starts not working into these cells with the date picker.

Shift Times					Calendar					Resources					User-defined				
		Date From		Date To		Reduce Time To		Comment											
1		2015/05/13				12:00 -24:00		Birthday of Boss											
2		2015/05/25						Memorial Day											
3		2015/07/03						Independence Day											
4		2015/07/13		2015/07/19				Retooling											
5		2015/09/07						Labor Day											

Remarks

Proceed as follows:

- Click the **Inheritance** check box so that it looks like this .
- Double-click into the text box and click the down arrow .
- Select a date in the calendar. Click the right or left buttons to move to another month.

To designate an entire day as a day-off, only select a start date, and no end date (**Date To**), and no **Reduce Time To**.

SimTalk

Calendar [SimTalk]

Date To

Type the date at which your plant stops not working into these cells with the date picker.

Shift Times		Calendar	Resources	User-defined	
	Date From	Date To	Reduce Time To	Comment	
1	2015/05/13		12:00 -24:00	Birthday of Boss	
2	2015/05/25			Memorial Day	
3	2015/07/03			Independence Day	
4	2015/07/13	2015/07/19		Retooling	
5	2015/09/07			Labor Day	

Remarks

Proceed as follows:

- Click the **Inheritance** check box so that it looks like this .
- Double-click into the text box and click the down arrow .

Select a date in the calendar. Click the right or left buttons to move to another month.

To designate a consecutive number of days-off, enter a start date (**Date From**), and an end date, and no **Reduce Time To**.

SimTalk

Calendar [SimTalk]

Reduce Time To

Type in the hour and the minute when the reduced **Working time** starts, a hyphen and the hour and minute the reduced time ends.

Shift Times				
Calendar				
Resources				
User-defined				
	Date From	Date To	Reduce Time To	Comment
1	2015/05/13		12:00 -24:00	Birthday of Boss
2	2015/05/25			Memorial Day
3	2015/07/03			Independence Day
4	2015/07/13	2015/07/19		Retooling
5	2015/09/07			Labor Day

Remarks

Note

Click the **Inheritance** check box so that it looks like this .

If your plant only works half a day on Christmas Eve, for example the shift from midnight to noon, you would type 0:00 - 12:00 into the cell **Reduce Time To**.

To designate several days on which your plant only works part of the time, type the start date in the cell in the column **Date From**, and the end date in the cell in the column **Date To**. Then, type the period of time in the cell **Reduce Time To**.

Note

The *ShiftCalendar* combines the reduced time and the definition of the shifts for a day. If, for example, the start time of a day with a reduced working time falls on a break, then this work day starts with a break.

SimTalk

Calendar [SimTalk]

See also

Working [state, material flow objects]

Comment [reduce time to]

Type a description of the event causing the **Reduce Time To** in this cell.

Note

Click the **Inheritance** check box so that it looks like this .

See also

Reduce Time To

Tab Resources

Tab Resources [ShiftCalendar]

Assign resources, meaning objects, to the *ShiftCalendar* on the tab **Resources**.

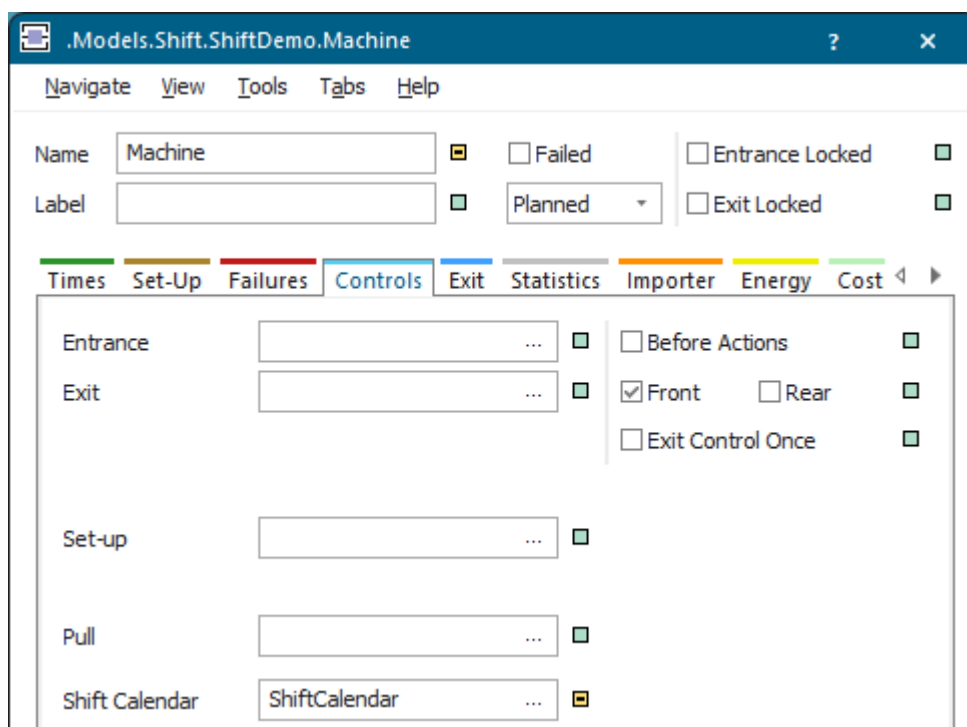
Remarks

A resource is any of the built-in **Material Flow Objects**, or a *Frame* in which you modeled a machine, whose **Working** hours you would like to control with the *ShiftCalendar*.

To add a resource to the tab **Resources**, drag it from the *Frame* window over the icon of the *ShiftCalendar* and drop it there. You can also select and then drag-and-drop multiple objects at the same time!

Shift Times	Calendar	Resources	User-defined
		Objects	
1	SubFrame		
2	Machine		

This automatically enters the *ShiftCalendar* into the text box **Shift Calendar** on the tab **Controls** of the material flow object.



To manipulate the contents of the list, use the commands on the [Context Menu of Embedded Lists](#).

See also

[Specify the Stations Which the ShiftCalendar Controls](#)

[Resources \[SimTalk\] - ShiftCalendar](#)

[Shift Calendar \[tab Controls\]](#)

Objects [ShiftCalendar]

Assign resources to the *ShiftCalendar* on the tab **Resources**.

Remarks

A resource is any of the built-in **Material Flow Objects**, or a *Frame* in which you modeled a machine, whose **Working** hours you would like to control with the *ShiftCalendar*.

Shift Times	Calendar	Resources	User-defined
Objects			
1	SubFrame		
2	Machine		

- Type the path to and the name of the resource object in a cell. Or
- Drag the resource object to the icon of the *ShiftCalendar* and drop it there. Or
- Type the name of the *ShiftCalendar* object on the tab **Controls** of the material flow object into the text box **Shift Calendar**.

Note

Plant Simulation pauses all resources you assign to a shift.

SimTalk

Resources [SimTalk] - ShiftCalendar

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

File Menu

File Menu [ShiftCalendar]

The **File** menu provides the commands listed in the table of contents to the left.

Import Shift Times

Imports the times during which your plant works in shifts from a tab delimited text file.

Export Shift Times

Exports the shift times you defined on the **Tab Shift Times** to a file.

Remarks

You can then import this file into another *ShiftCalendar* or edit in a word processing program.

See also

[Tab Shift Times](#)

Import Calendar

Imports the calendar, i.e., the days during which your plant works in shifts from a tab delimited text file.

Export Calendar

Exports the shift calendar you defined on the **Tab Calendar** to a file.

Remarks

You can then import this file into another *ShiftCalendar* or edit in a word processing program.

See also

[Tab Calendar](#)

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

View Menu

The **View Menu** provides commands to access its functions.

[Refresh \[on View menu\]](#)

[Show Assigned Objects \[ShiftCalendar\]](#)

[Show Attributes and Methods \[on View menu\]](#)

SimTalk

[updateDialog \[SimTalk\]](#)

Show Assigned Objects [ShiftCalendar]

Selects the objects which are assigned to the *ShiftCalendar* as resources in the *Frame* window.

Instead, you can also select [Show Assigned Objects \[in Frame\]](#) on the context menu of the *ShiftCalendar* in the *Frame*.

Tools Menu

The **Tools Menu** provides these menu commands:

Edit Controls

Edit Observers

Help Menu

The commands are described under the [Help Menu](#).

Methods of the ShiftCalendar

_Methods of the ShiftCalendar

The *ShiftCalendar* provides:

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			((Contents:table)) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).